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(54) **PERSONALIZED FALL DETECTOR**

(56) **References Cited**

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U.S. PATENT DOCUMENTS

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10,980,491 B1 4/2021 Jones et al.  
12,027,035 B2 \* 7/2024 Ten Kate ..... G08B 29/185  
(Continued)

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FOREIGN PATENT DOCUMENTS

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CN 106407996 A 2/2017  
CN 109166275 A 1/2019  
(Continued)

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OTHER PUBLICATIONS

(21) Appl. No.: **18/746,880**

International Preliminary Report on Patentability, relating to Applicaiton  
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**Related U.S. Application Data**

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(57) **ABSTRACT**

(30) **Foreign Application Priority Data**

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A method and system for training a fall detection classifier using subject-specific movement data. Movement data responsive to a subject's movements during everyday activities are obtained over a predetermined data collection period. For each detected event in the movement data, values for one or more parameters that may (together or individually) indicate a fall are obtained. The obtained values are used to generate a subject-specific probability distribution for non-fall events. A fall event probability distribution is obtained using the non-fall event probability distribution. This fall event probability distribution can then be subsequently processed, with reference to a threshold value, to distinguish fall events from non-fall events.

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**G01P 13/00** (2006.01)

**G08B 21/04** (2006.01)

(52) **U.S. Cl.**

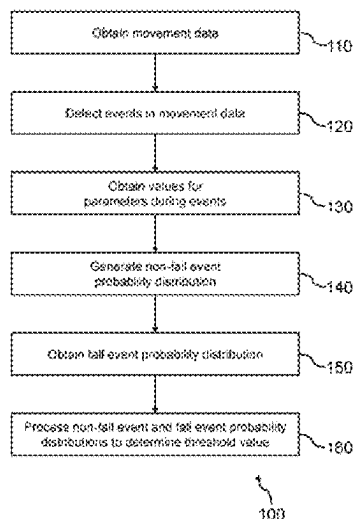
CPC ..... **G08B 29/185** (2013.01); **G01P 13/00**  
(2013.01); **G08B 21/0446** (2013.01)

(58) **Field of Classification Search**

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21/0446; G08B 29/26; G01P 13/00

See application file for complete search history.

**20 Claims, 4 Drawing Sheets**



(56)

**References Cited**

## U.S. PATENT DOCUMENTS

|              |      |         |                 |                           |
|--------------|------|---------|-----------------|---------------------------|
| 2006/0214806 | A1   | 9/2006  | Clifford et al. |                           |
| 2009/0224925 | A1 * | 9/2009  | Gannot .....    | G08B 21/0446<br>340/573.1 |
| 2012/0106782 | A1   | 5/2012  | Nathan et al.   |                           |
| 2013/0138395 | A1 * | 5/2013  | Baggen .....    | G08B 21/043<br>702/181    |
| 2018/0174420 | A1   | 6/2018  | Clark et al.    |                           |
| 2018/0336773 | A1   | 11/2018 | Hanson et al.   |                           |
| 2019/0175074 | A1   | 6/2019  | Zhang et al.    |                           |

## FOREIGN PATENT DOCUMENTS

|    |              |    |        |
|----|--------------|----|--------|
| DE | 102008049194 | A1 | 4/2010 |
| WO | 2009090584   | A2 | 7/2009 |

## OTHER PUBLICATIONS

International Preliminary Report on Patentability, relating to Applicaiton No. PCT/EP2020/083660, dated Jun. 9, 2022.

Zhang, T. et al., "Fall Detection by Wearable Sensor and One-Class SVM Algorithm", ICIC 2006, LNCIS 345, pp. 858 363, 2006.

Yu, M. et al., "Fall Detection for the Elderly in a Smart Room by Using an Enhanced One Class Support Vectro Machine", 2011, Advanced Signal Processing Group, Electronic and Electrical Engineering Department, Loughborough University, Loughborough, Leicester, UK.

Pierleoni, P. et al., "A High Reliability Wearable Device for Elderly Fall Detection", IEEE Sensors Journal, vol. 15, No. 8, Aug. 2015. USPTO. Office Action relating to U.S. Appl. No. 16/953,487, dated Sep. 7, 2023.

\* cited by examiner

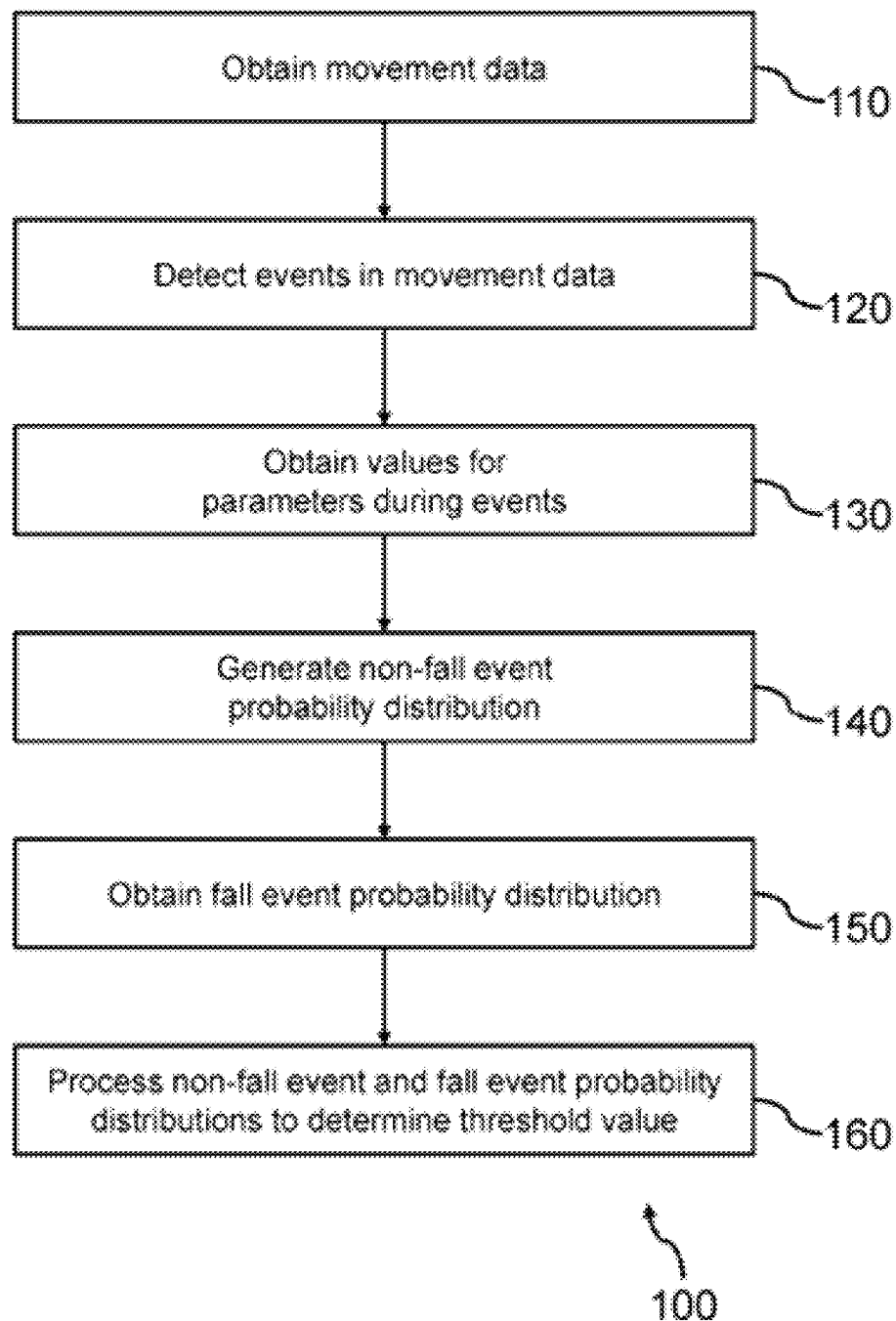


FIG. 1

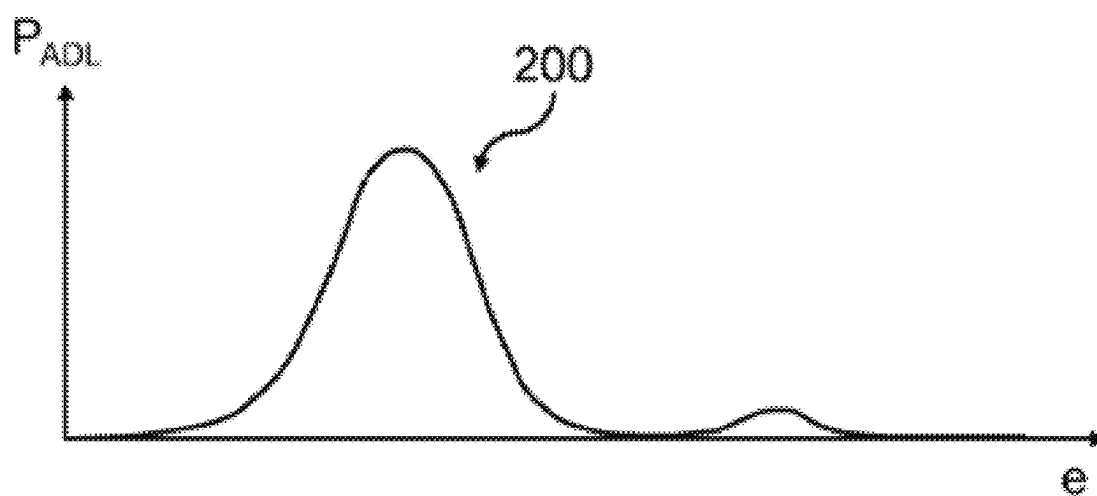


FIG. 2

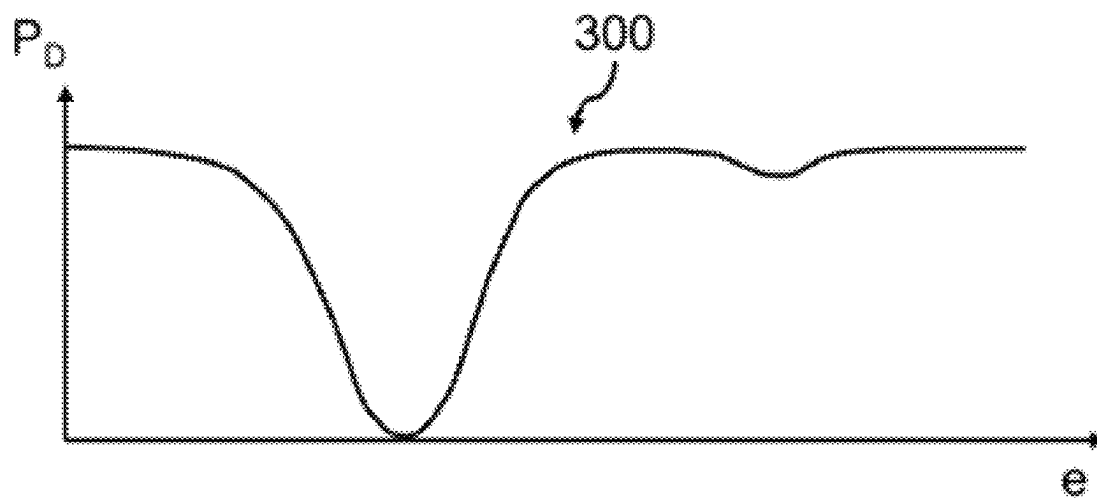


FIG. 3

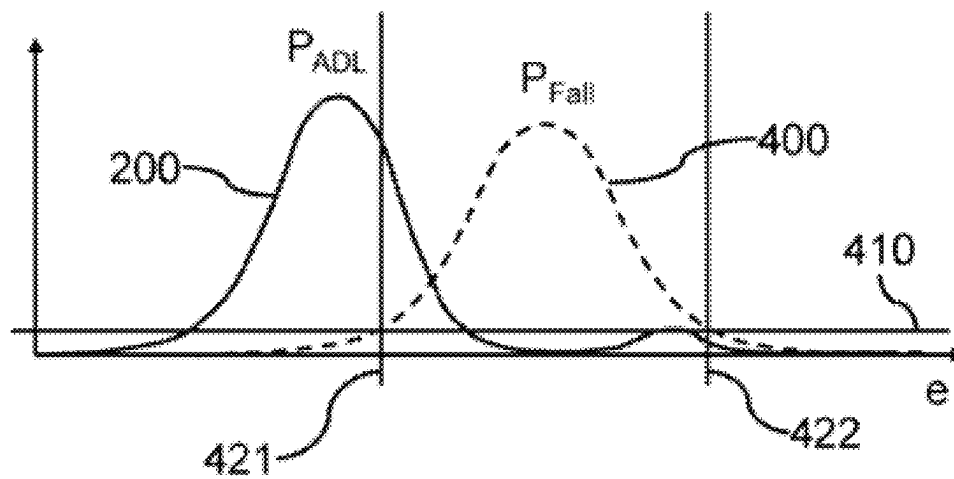


FIG. 4

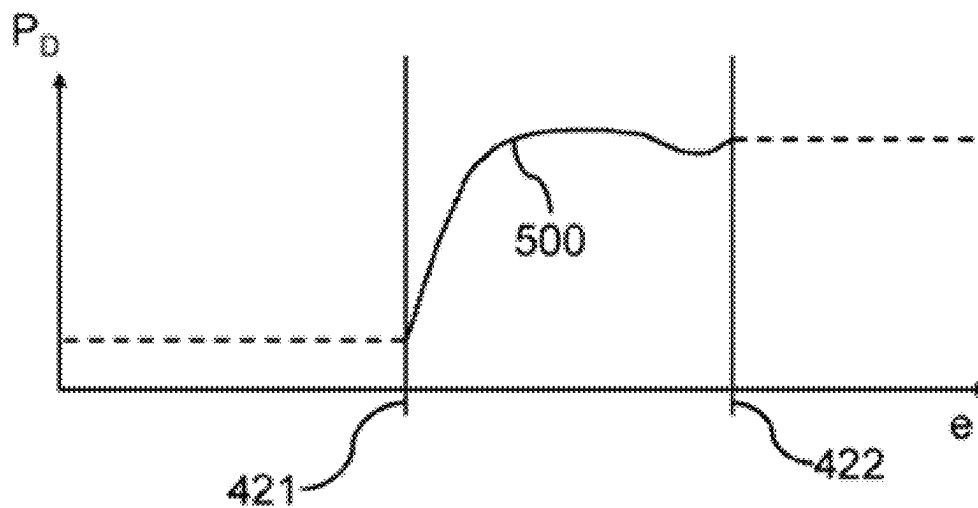
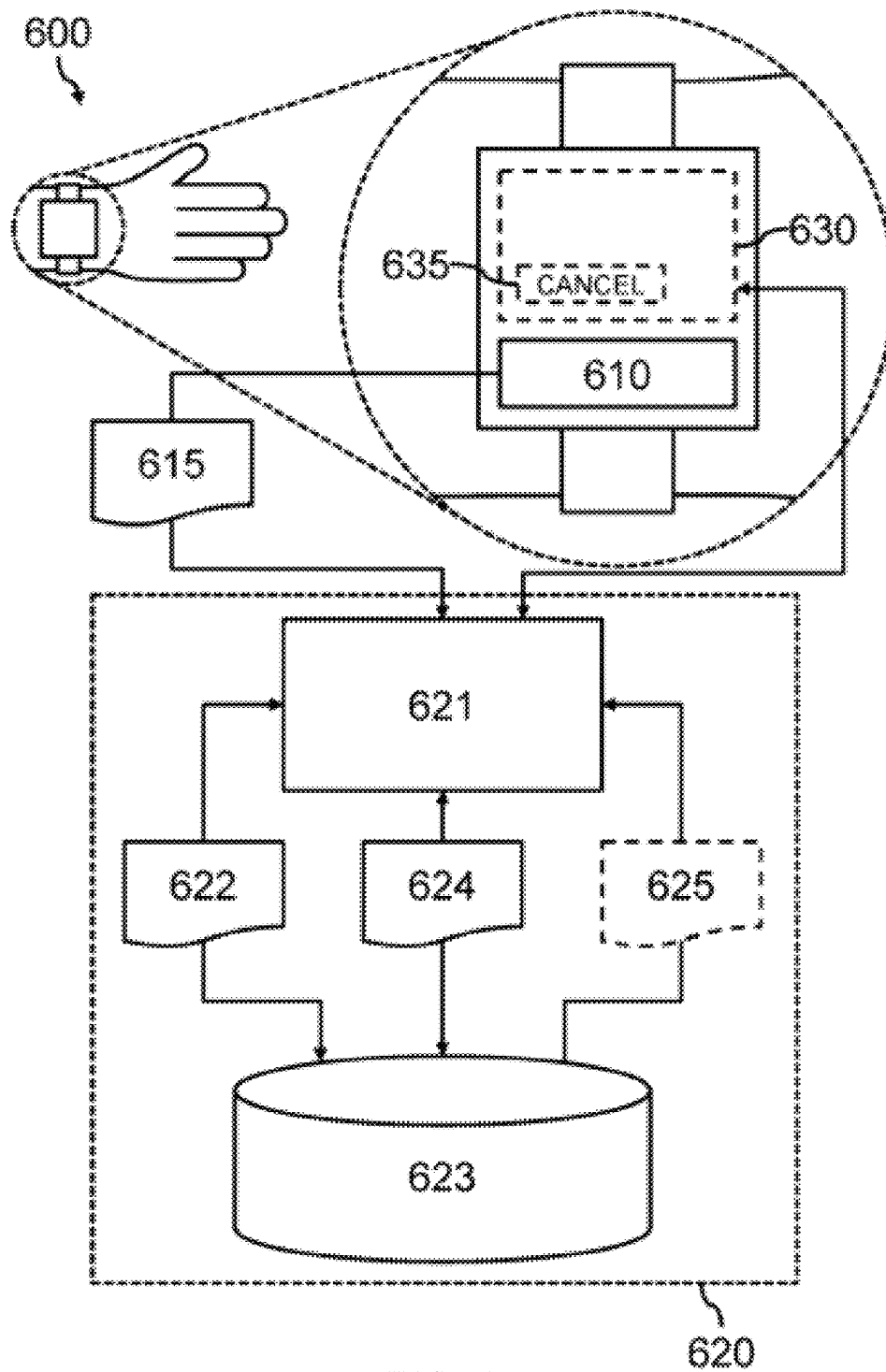


FIG. 5



## PERSONALIZED FALL DETECTOR

## CROSS REFERENCE TO RELATED APPLICATIONS

This U.S. patent application is a continuation of, and claims priority under 35 U.S.C. § 120 from, U.S. patent application Ser. No. 16/953,487, filed on Nov. 20, 2020, which claims priority under 35 U.S.C. § 119(a) to European Patent Application No. 1921267.2, filed on Nov. 29, 2019. The disclosures of these prior applications are considered part of the disclosure of this application and are hereby incorporated by reference in their entireties.

## TECHNICAL FIELD

This disclosure relates to fall detection in Personal Emergency Response Systems.

## BACKGROUND

Personal Emergency Response Systems (PERS) enable elderly and disabled people to live independently by summoning immediate help when an adverse event, such as a fall, occurs. The use of such systems allows people who would otherwise require round-the-clock care to live in their own homes, and reduces their care costs.

Some PERS systems rely on subject input, such as a button, to raise an alarm when assistance is required, but this means that a subject is unable to access immediate medical help in the case of the subject's becoming unconscious as a result of a fall, when such help is most critical. There is also a risk that the subject may fall in such a way that, while remaining conscious, he/she is unable to activate the alarm.

For these reasons, automatic fall detectors have been developed that can summon help without needing subject input. These detectors generally come in the form of wearable devices that contain sensors to monitor subjects' movements and processors that decide whether a detected movement is the result of a fall. These devices are often located around the neck, at the waist or around the wrist, but other locations are also conceivable, including such locations as ears (for example, in hearing aids). Wrist-located devices are becoming increasingly popular as smartwatches, on which fall detection apps may be installed, become more widely used.

Current automatic fall detectors are not able to distinguish with complete accuracy between falls and movements that occur during everyday activities. Fall detection accuracy is particularly low in wrist-located devices.

In order to avoid failing to detect genuine falls, fall detectors are configured to have a sufficiently high false alarm rate (the rate of non-fall events classified as fall events) to minimize the number of falls that are not detected as such. A high false alarm rate necessitates the use of a cancel function to avoid summoning help needlessly; subjects can activate this function when the fall detector incorrectly detects a fall. However, some subjects may find the cancel function difficult to use, become panicked when a false alarm occurs and forget to activate the cancel function, or accidentally activate the cancel function when a genuine fall has occurred.

There is therefore a need for a fall detector with improved accuracy, such that false alarms are unlikely enough that a cancel function is not required.

## SUMMARY

The disclosure is defined by the claims.

According to examples in accordance with an aspect of the disclosure, there is provided a computer-based method of training a classifier to distinguish between a fall event and a non-fall event of a subject.

The method includes: obtaining movement data responsive to the subject's movement during everyday activities over a predetermined data collection period; detecting one or more events in the movement data; obtaining a value for each of a one or more parameters from the movement data at the time of the event for each event that occurs during the predetermined data collection period; generating a non-fall event probability distribution by calculating a probability distribution for a combination of the one or more parameters using the obtained values for each of the one or more parameters, wherein the combination of the one or more parameters is capable of indicating the probability of a fall; obtaining a fall event probability distribution using the non-fall event probability distribution; and obtaining a threshold value for the fall event probability distribution that distinguishes between a fall event and a non-fall event.

This method uses a subject-specific probability distribution that an event is not a fall in order to distinguish between a fall and a non-fall event with greater accuracy than methods that use a probability distribution that an event is (not a) fall based on a population average, as different subjects will exhibit different typical movement behaviors.

The subject's movements are detected during everyday activities over an initial data collection period; events are recorded based on the movement data, and one or more values for parameters that could affect the probability that an event is a fall, such as height change and orientation, are computed. Once the initial data collection period is completed, for example by a time interval having been passed or by a minimum number of events being collected, these values are used to determine a subject-specific probability distribution for non-fall events.

The subject-specific probability distribution for non-fall events is used to obtain or derive a fall event probability distribution.

In subsequent processing for detection of a fall event, a calculated fall event probability for the combination of one or more values (for the one or more parameters) for a detected event can be compared with a threshold value in order to decide whether the detected event is a fall or a non-fall event.

Events may be detected by setting at least one of the one or more parameters as a trigger parameter and defining an event as occurring at a time at which a value of the at least one trigger parameter exceeds or falls below a corresponding predetermined trigger threshold value. Other methods of detecting an event will be apparent to the skilled person, e.g. by processing a chunk of movement data using a machine-learning method.

The proposed examples enable a fall detector to be personalized to the movement of a subject, i.e. to better distinguish between a fall and a non-fall of the subject. In particular, a fall detection probability distribution (that is used to distinguish between a fall and a non-fall) is personalized to a user. This reduces the likelihood of false alarms being generated when a subject has not actually fallen.

In some examples, the one or more parameters includes a plurality of parameters, so that the combination of one or more parameters includes a combination of a plurality of parameters.

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In some examples, the calculated probability distribution for the combination of the one or more parameters is calculated by determining a probability distribution for each of the one or more parameters, and defining the calculated probability distribution for the combination of the one or more parameters as the product of the determined probability distributions. Put another way, the logarithm of the probability distribution (in an arbitrary base) for the combination of the one or more parameters is computed as the sum of the logarithms of the determined individual probability distributions (in the same base).

The step of obtaining a fall event probability distribution using the non-fall event probability distribution may include calculating the fall event probability distribution using the non-fall event probability distribution. The step of processing the non-fall probability distribution to obtain the fall event probability distribution may include processing the non-fall event probability distribution such that there is a high fall event probability for values at which the non-fall event probability is low and vice versa. For example, the fall event probability distribution may be the inverse or complement of the non-fall event probability distribution.

In another example, the step of obtaining the fall event probability distribution may include dividing a predetermined estimation of the fall event probability distribution by the non-fall event probability distribution. The quotient may be normalized to derive a fall event probability distribution, or may by itself effectively form a fall event probability distribution.

Thus, a probability distribution for fall events is generated by using the idea that events that are likely to be non-fall events are unlikely to be fall events. This avoids the need for directly obtaining reliable fall data of a user, which are difficult and time-consuming to obtain.

In some examples, the step of obtaining the fall event probability distribution includes processing only part of the non-fall event probability distribution, which part of the non-fall probability distribution to be processed being determined using a predetermined estimation of a fall event probability distribution.

This recognizes that there may be some values for the combination of values of the one or more parameters for an event at which the non-fall event probability is low and at which it is unlikely that the event is a fall, according to an estimated fall event probability distribution. Using an estimated fall event distribution to limit the part of the non-fall event probability distribution that is processed to obtain a fall event probability distribution excludes such values, which reduces the number of false alarms detected, that is, the number of non-fall events that are determined to be falls.

The part of the non-fall event probability distribution processed to obtain the fall event probability distribution may include the non-fall event probability distribution for values of the combination of the one or more parameters for which the predetermined estimation of a fall event probability exceeds a predetermined minimum value.

The part of the non-fall event probability distribution may include the non-fall event probability distribution for values of the combination of the one or more parameters lying between the combination of the one or more parameters having a maximum value in the non-fall event probability distribution and the combination of the one or more parameters having a maximum value in the predetermined estimation of a fall event probability distribution.

The fall event probability distributed may be calculated by computing a likelihood ratio, defined by dividing the

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predetermined estimation of a fall event probability distribution by the determined non-fall event probability distribution.

In this way, the likelihood ratio test is applied in order to determine the fall event probability distribution. The likelihood ratio test has been shown, by the Neyman-Pearson theorem, to be the most powerful test for determining whether or not an event is a fall for a given false alarm rate. The most powerful test is the test with the lowest probability of determining a genuine fall event to be a non-fall event.

The threshold value for the fall-event probability distribution may be computed using the non-fall event probability distribution and the fall event probability distribution. In other examples, the threshold value may be set as a predetermined value.

In other examples, the threshold value may be set as the value at which the false alarm rate for the subject, defined as the rate at which non-fall events are determined to be fall events, is equal to a predetermined rate. In this way, the threshold value may be personalized to the subject to ensure that the false alarm rate for that subject is at an acceptable level.

In other examples, the threshold value is a predetermined threshold value, e.g. one that may have been used if a population-based fall event probability distribution were used.

According to examples in accordance with an aspect of the disclosure, there is provided a computer program including code means for implementing the method of any previously described method when said program is run on a processing system.

According to a concept of the disclosure, there is provided a processing system adapted to: obtain movement data responsive to the subject's movement during everyday activities over a predetermined data collection period; detect one or more events in the movement data; obtain a value for each of one or more parameters from the movement data at the time of the event for each event that occurs during the predetermined data collection period; a non-fall event probability distribution by calculating a probability distribution for a combination of one or more parameters using the obtained values for each of one or more parameters, wherein the combination of the one or more parameters is capable of indicating the probability of a fall; obtain a fall event probability distribution using the non-fall event probability distribution; and obtain a threshold value for the fall event probability distribution that distinguishes between a fall event and a non-fall event.

The processing system may be adapted to obtain the fall event probability distribution by processing the non-fall event probability distribution, such that there is a high fall event probability for values at which the non-fall event probability is low and vice versa.

There is also proposed a system for detecting a fall of a subject, including one or more sensors for obtaining movement data responsive to the subject's movement, and the processing system described above, further configured to: receive the movement data from the one or more sensors; detect events in the movement data after the predetermined data collection period has elapsed; and classify each detected event as a fall event or a non-fall event by comparing the fall event probability for the combination of obtained values for the one or more parameters, for the event, with the threshold value.

The processing system may, when the threshold value corresponds to a predetermined false alarm rate, be further configured to determine a false alarm probability, defined as



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the quotient between the predetermined false alarm rate and a rate at which events are detected for the subject, and the system for detecting a fall of a subject may further include a user interface configured to provide a cancel function, which, when activated by the subject, instructs the processing system to re-classify a detected fall event as a non-fall event, wherein the user interface is configured to display the predetermined false alarm rate and/or the false alarm probability, and optionally to receive a user instruction to disable the cancel function, and the user interface is configured to selectively disable the cancel function, after the predetermined data collection period has elapsed, responsive to the value of the determined false alarm probability and/or a user instruction.

Disabling the cancel function when the system is judged to classify events with sufficient accuracy reduces interaction complexity and increases subject safety by preventing the cancel function from being accidentally activated in the case of a real fall event.

These and other aspects of the disclosure will be apparent from and elucidated with reference to the examples described hereinafter.

## DESCRIPTION OF DRAWINGS

For a better understanding of the disclosure, and to show more clearly how it may be carried into effect, reference will now be made, by way of example only, to the accompanying drawings, in which:

FIG. 1 is a flow diagram of a method of training a classifier to distinguish between a fall event and a non-fall event of a subject, according to an implementation of the disclosure.

FIG. 2 is a graph showing an example of a non-fall event probability distribution.

FIG. 3 is a graph showing an example of a fall event probability distribution generated using a method according to an implementation of the disclosure.

FIG. 4 is a graph showing an example of a non-fall event probability distribution and an example of a predetermined estimation of a fall event probability distribution.

FIG. 5 is a graph showing another example of a fall event probability distribution generated using a method according to an implementation of the disclosure.

FIG. 6 is a diagram of a system for detecting a fall of a subject according to the disclosure.

## DETAILED DESCRIPTION

The disclosure will be described with reference to the Figures.

It should be understood that the detailed description and specific examples, while indicating exemplary implementations of the systems and methods, are intended for the purposes of illustration only and are not intended to limit the scope of the disclosure. These and other features, aspects and advantages of the systems and methods of the present disclosure will become better understood from the following description, appended claims and accompanying drawings. It should be understood that the same reference numerals are used throughout the Figures to indicate the same or similar parts.

Existing fall detection systems typically consist of two phases. In a first phase, values for parameters that may indicate the likelihood of a fall are obtained for an event from data from sensors such as accelerometers and barometric sensors. Other sensors may include gyroscopes, mag-

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netometers, ultrasound sensors, radar sensors, skin conductivity sensors and pulse (PPG) sensors, for example. The parameters may include any of height, height change, orientation, orientation change, velocity, duration and impact (distance, physiological quantities). Events are typically defined using a trigger, which monitors at least one trigger parameter: an event is considered to have occurred at the point at which the value of a trigger parameter exceeds or falls below a corresponding predetermined trigger threshold value (or when a change of a value is greater than a predetermined change). For example, an event may be detected when the barometer indicates a height drop of more than 50 cm, or the accelerometer indicates an acceleration of more than 2 g.

In a second phase, a classifier (e.g. a processing device) takes one or more of the obtained parameter values from the event and outputs a value indicating or predicting whether or not the event is a fall. In some cases, the obtained parameter value(s) are used by the classifier to calculate a probability that a fall has occurred, e.g. using a fall event probability distribution, and the classifier applies a threshold to the calculated probability to decide whether that probability is predict, with a certain degree of accuracy, whether or not a fall has occurred.

According to a concept of the disclosure, there is proposed a method and system for training a fall detection classifier using subject-specific movement data. Movement data responsive to a subject's movements during everyday activities are obtained over a predetermined data collection period. For each detected event in the movement data, values for one or more parameters that may (together or individually) indicate a fall are obtained. The obtained values are used to generate a subject-specific probability distribution for non-fall events. A fall event probability distribution is obtained using the non-fall event probability distribution. This fall event probability distribution can then be subsequently processed, with reference to a threshold value, to distinguish fall events from non-fall events.

Implementations/examples are at least partly based on the realization that different people exhibit different movement behaviors, and that a classifier trained on subject-specific movement data can detect a subject's falls with greater accuracy than a classifier trained using population data.

Illustrative implementations/examples may, for example, be employed in fall detectors and/or fall detection systems in PERS systems at a subject's home or in specialized accommodation such as Senior Living Facilities, in hospital products, and/or using apps designed for smartwatches.

In any described implementation/example, a probability distribution may be a joint probability distribution or a multivariate probability distribution. This can be expressed as a cumulative distribution function or a joint probability density function.

FIG. 1 illustrates a method 100 for training a classifier to distinguish between a fall event and a non-fall event of a subject, according to an implementation of the disclosure.

The method 100 begins with step 110, in which movement data responsive to the subject's movement during everyday activities, e.g. activities of daily living (ADL), are obtained over a predetermined data collection period.

The movement data includes sufficient information for a one or more predefined parameters to be measured from the movement data, wherein each parameter, or a combination of the parameters, is capable of indicating the probability of a fall. Examples of parameters capable of indicating the

probability of a fall include height, height change, orientation, orientation change, air pressure change, velocity, acceleration, duration and impact.

The movement data may include data from at least one suitable sensor. Examples of suitable sensors include accelerometers, barometric sensors, tilt switches, vibration sensors, gyroscopes, magnetometers, ultrasound sensors, radar sensors, skin conductivity sensors and pulse (photoplethysmography, PPG) sensors. These data are collected as the subject carries out everyday activities, such as standing up, sitting down, lying down, bending over, walking and climbing stairs. For wrist-located devices, the movements also include the many motions related to activities such as pointing, waving, gesturing (during conversations), (car) driving, washing dishes, doing laundry, and (vacuum) cleaning.

The predetermined data collection period may be a week, one or a few days, a month or some other suitable length of time. An initial short data collection period, for example, a day, may be used to obtain some movement data, then data may continue to be collected over a longer data collection period to refine the results.

The predetermined data collection period may also be the length of time taken to detect a predetermined number of events, or the data collection period may be defined by both a minimum number of events and a minimum duration. Thus, in some examples, the predetermined data collection period may be defined as a period during which a predetermined number of events (described below) are detected. Other definitions are also conceivable, including user controlled methods.

The subject's movement data may be kept up-to-date, ensuring that fall events continue to be accurately distinguished from non-fall events even if the subject's typical movement behaviors change over time, by repeating the method 100 at regular intervals, or by continually collecting movement data and using data from a moving window, for example, data from the last week, in the remaining steps of the method 100.

At step 120, the method includes detecting one or more events in the movement data. An event may be defined by setting at least one of the predefined parameter(s), or a combination of parameters, as a trigger parameter and defining an event as occurring at a time at which a value of the at least one trigger parameter exceeds or falls below a corresponding predetermined trigger threshold value. The used parameter(s) is (are) usable again in the next steps, but this is not essential.

The at least one parameter to be set as a trigger parameter may be predefined. The at least one parameter may be a parameter that research shows may be configured to detect all, or almost all, fall events while detecting a minimal number of non-fall events. The at least one trigger parameter may be selected based on the computational complexity required to obtain a value of the trigger parameter. Examples of trigger parameters may include air pressure change and acceleration size.

A suitable trigger threshold may be determined from literature, or from simulations of falls, and may be set to be a value for which all, or almost all, fall events will pass. For example, an event may be detected when the barometer indicates a height drop of more than 50 cm, or the accelerometer indicates an acceleration of more than 2 g.

Other methods of detecting events, such as the use of a machine learning algorithm to directly process chunks of movement data, will be apparent to the skilled person.

At step 130, the method includes obtaining a value for each of one or more parameters at the time of the event for each event that occurs during the predetermined data collection period. The one or more parameters preferably includes a plurality of parameters. This data may be labelled "event information".

In some examples, values are obtained for all of the predefined parameters; in other examples, a selection process may be used to select a subset of a plurality of predefined parameters that yields best accuracy for the subject. Examples of methods for selecting parameters will be apparent to the skilled person, known in the art as "feature selection", and may include forward search, backwards search and LASSO.

At step 140, the method includes generating a non-fall event probability distribution ( $P_{ADL}$ ) by calculating a probability distribution for a combination of the one or more parameters using the event information, i.e. using the obtained values for each of the one or more parameters from all the events that occur during the predetermined data collection period.

This probability distribution may be calculated by determining a probability distribution for each of the one or more parameters, and defining the calculated probability distribution for the combination of the one or more parameters as the product of the determined probability distributions.

The log of the probability distribution for the combination of the one or more parameters may therefore be found by calculating the sum of the logs of the determined probability distributions.

Extreme values, due to, for example, the probability distribution for a parameter being close to zero, may be clipped.

In other examples, the non-fall event probability distribution may be calculated by first combining two or more parameters (e.g. into subsets of parameters), before calculating the non-fall event probability distribution for the combination.

The non-fall event probability distribution may be multivariate, i.e. form a joint probability distribution.

The probability distribution of a single parameter may be found by curve-fitting to the collected set of values of that parameter and normalizing the curve's integral to unit size. In some examples, all collected values for a parameter may be used to generate the probability distribution for the parameter, operating under the assumption that the number of falls that occur during the predetermined data collection period will be low enough not to significantly affect the calculated probability distribution. In other examples, outliers may be removed before the curve-fitting, by, for example, excluding values from events that have a high probability of being fall events, according to some predetermined estimation of a fall event probability, and/or values from events that are defined as fall events based on subject input.

Since the non-fall event probability distribution is generated from subject-specific movement data rather than a population average, it may be used to distinguish between fall events and non-fall events with greater accuracy.

In some examples, a plurality of non-fall event probability distributions (for different combinations of one or more parameters) may be generated from the movement data. The plurality of non-fall probability distributions may be employed in subsequent steps in place of a single non-fall probability distribution.

At step 150, a fall event probability distribution ( $P_D$ ) for the combination of the one or more parameters is obtained

(i.e. those parameters contained in the event information) using the non-fall event probability distribution.

Since the number of falls during the predetermined data collection period may be low, a fall event probability distribution cannot be reliably obtained using or incorporating real, subject-specific fall data during the data collection period. That being said, in some examples, real subject-specific data from any falls that are detected after the predetermined data collection period is complete may be used to further refine the fall event probability distribution.

In preferred examples, the fall event probability distribution is calculated by processing the non-fall event probability distribution. This process may follow the approach of anomaly detection, in which events outside the non-fall event probability distribution (anomalies) are classified as fall events.

A first example method of obtaining the fall event probability distribution may include computing a likelihood ratio. This process may include dividing a predetermined estimation of a fall event probability distribution by the non-fall event probability distribution generated from the subject's movement data. The likelihood ratio may then be normalized to formulate a probability distribution. Of course, in some examples, the likelihood ratio may itself act as a probability distribution (albeit

Another example method of obtaining a fall event probability distribution includes processing the non-fall event probability distribution such that there is a high fall event probability for values at which the non-fall event probability is low and vice versa. A high probability may be considered to be a probability above 0.5, where the probability is on a scale of 0 to 1, while a low probability may be considered to be a probability below 0.5. These are purely exemplary values, and the skilled person might consider other threshold values for determining whether a probability is "high" and "low".

There are a number of ways in which the fall event probability distribution may be defined in terms of the non-fall probability distribution such that there is a high fall event probability for values at which the non-fall event probability is low and vice versa. For example, the fall event probability distribution may be defined as the inverse/reciprocal of the non-fall event probability distribution, the probability complement of the non-fall event probability distribution (calculated by performed 1-the non-fall event probability distribution), or the probability complement of the non-fall event probability distribution divided by the non-fall event probability distribution.

In other examples, the fall event probability distribution is obtained by selecting a predetermined fall event probability distribution (e.g. from a number of options) based upon the non-fall event probability distribution. This may include, for example, consulting a database correlating predetermined fall event probability distributions to predetermined non-fall event probability distributions, and selecting the fall event probability distribution that corresponds to a predetermined estimation of a probability distribution that most closely matches (e.g. calculated using a cross-correlation technique or the like) the generated non-fall event probability distribution.

Other methods of obtaining a fall event probability distribution using a generated non-fall event probability distribution will be apparent to the skilled person, the selection of which depends upon preferred implementation details. Further examples are given in FIGS. 2 to 5.

FIG. 2 illustrates an example non-fall event probability distribution 200, also labelled  $P_{ADL}$ , generated from a sub-

ject's movement data over the predetermined data collection period, as described in step 140 of method 100.

FIG. 3 illustrates an example fall event probability distribution 300, also labelled  $P_D$ , obtained by taking the complement of the non-fall event probability distribution 200. The fall event probability distribution 300 has a high probability where the non-fall event probability distribution 200 has a low probability, and a low probability where the non-fall event probability distribution 200 has a high probability.

The accuracy of a fall event probability distribution obtained by processing a non-fall event probability distribution may be improved by recognizing that there are some values for which the non-fall event probability is low that are unlikely to correspond to fall events.

FIG. 4 illustrates the example non-fall event probability distribution 200, also labelled  $P_{ADL}$ , alongside an example of a predetermined estimation of a fall event probability distribution 400, also labelled  $P_{FALL}$ . The predetermined estimation 400 may be generated using simulated fall data, or by some other method, such as from literature or historic measurements. The predetermined estimation may be obtained, for example, from a database, storage or other memory module.

As can be seen from FIG. 4, both the non-fall event probability distribution 200  $P_{ADL}$  and the predetermined estimation of a fall event probability distribution 400,  $P_{FALL}$  have a low probability at very low and very high values of the combination of the one or more parameters (plotted along the x-axis). Referring back to FIG. 3, the fall event probability distribution 300, obtained by reverting (i.e. computing the complement) the non-fall event probability distribution 200, has a high probability at these values.

In an example, only part of the non-fall event probability distribution is processed to obtain the fall event probability distribution, in order to exclude parts of the non-fall event probability distribution where a predetermined estimation of a fall event probability distribution has a low probability. Which part of the non-fall event probability distribution is processed is determined using the predetermined estimation of a fall event probability distribution.

FIG. 4 shows an example of how the part of the non-fall event probability distribution to be processed may be determined. In this example, a minimum value 410 for the predetermined estimation of a fall event probability distribution 400 is chosen. The minimum value 410 may be a percentage of the maximum probability of the predetermined estimation of a fall event probability distribution, or may be a predetermined value. The non-fall event probability distribution 200 is only processed for values of the combination of the one or more parameters for which the predetermined estimation of a fall event probability exceeds the minimum value 410.

In FIG. 4, the part of the non-fall event probability distribution 200 that is processed is the part between the boundaries 421 and 422. There are two boundaries 421 and 422 in FIG. 4, but more complex probability distribution in which more boundaries are required may be used.

FIG. 5 illustrates an example of a fall event probability distribution 500 that may be obtained using this method. Between the boundaries 421 and 422, the fall event probability distribution 500 is obtained by taking the complement of the part of the non-fall event probability distribution 200 that lies between the boundaries 421 and 422.

The part of the fall event probability distribution 500 to the left of (lower) boundary 421 (i.e. having a lower value than the boundary 421) may be set as a constant equal to the

fall event probability at boundary **421**, or may be given some predetermined value. A suitable predetermined value may be the minimum value **410** used to determine which part of the non-fall event probability distribution is processed, although other values are contemplated.

Similarly, the part of the fall event probability distribution **500** to the right of (upper) boundary **422** (i.e. having a higher value than the boundary **422**) may be set as a constant equal to the fall event probability at boundary **422**, or given some predetermined value.

To save memory space, the fall event probability distribution may be simply stored as values between the lower **421** and upper **422** boundaries. During later processing, if an (x-axis) value lower than the lower boundary **421** is used, the corresponding probability may be clipped to the value at the lower boundary. Similarly, if an (x-axis) value higher than the upper boundary is used, the corresponding probability may be clipped to the value at the upper boundary **422**.

Other methods to generate the part of the fall event probability distribution that is outside the boundaries of **421**, **422** (i.e. define the probability values outside of the boundaries **421**, **422**) may be used. For example, the part of the predetermined estimation of a fall event probability distribution **400** outside the boundaries **421** and **422** may be used as the part of the fall event probability distribution outside the boundaries **421** and **422**.

Other methods may be used to determine which part of the non-fall event probability distribution is processed. For example, the part of the non-fall event probability distribution may include the part of the non-fall event probability distribution for values of the combination of the one or more parameters lying between the combination of the one or more parameters having a maximum value in the non-fall event probability distribution and the combination of the one or more parameters having a maximum value in the predetermined estimation of a fall event probability distribution.

In implementations in which a plurality of non-fall event probability distributions are generated in step **140**, any previously described method of generating a fall event probability distribution could be applied to each of the plurality of non-fall event probability distributions (thereby generating intermediate fall event probability distributions, one for each non-fall event probability distribution). The intermediate fall event probability distributions may be combined, e.g. by calculating the product or summing logs, in order to generate the (overall) fall event probability distribution.

Whilst presently illustrated using only two dimensions for the sake of improved clarity, in some examples, the non-fall event probability distribution and/or fall event probability distribution may be a joint probability density function, i.e. a multivariate probability distribution. This can be formatted using a plurality of two-dimensional probability distributions (e.g. for each parameter or sub-combination of parameters).

Different dimensions of the joint probability dimension function may therefore represent a different variable from the movement data. Dimensions of the multi-dimensional probability distribution may then be processed independently to generate a corresponding dimension for the fall-event probability distribution (e.g. by performing any previously described method for generating a fall-event probability distribution). In other words, each of a plurality of two-dimensional probability distributions forming the non-fall event probability distribution may be individually processed using a previously described method (e.g. invert-

ing or the like) to generate a two-dimensional probability distribution for the fall event probability distribution. This forms a multivariate probability distribution for the fall event probability distribution.

The two-dimensional probability distributions could be combined, e.g. using a product or summing logs, to generate a two-dimensional fall event probability distribution. Alternatively, the fall event probability distribution may be maintained as a joint probability density function.

Referring back to FIG. **1**, at step **160**, the method includes obtaining a threshold value, for the fall event probability distribution, that distinguishes between a fall event and a non-fall event.

Step **160** may include processing the non-fall event probability distribution and the fall event probability distribution in order to calculate the threshold value. This processing is performed using one or more computer-based algorithmic processes, examples of which are now provided.

In some examples, the threshold value may be chosen such that the classifier has a particular false alarm rate. This approach is particularly suited when the fall event probability distribution is calculated using a likelihood ratio. This method for determining the threshold value recognizes that classifiers based on the likelihood ratio test provide optimal detection accuracy, according to the Neyman-Pearson theorem. For a given false alarm rate, the likelihood ratio test provides the lowest probability of incorrectly classifying a fall event as a non-fall event.

In some implementations, the threshold value may be personalized by setting it at the fall event probability at which the false alarm rate for the subject is equal to a predetermined rate  $FA_{set}$ . The threshold value may be determined from the subject-specific non-fall event probability distribution, by using the fact that the integral of the non-fall event probability distribution over the region where the fall probability is above the threshold value equals the predetermined false alarm probability. This false alarm probability can be determined as the quotient between the predetermined false alarm rate  $FA_{set}$  and a trigger rate, where the trigger rate is defined as the number of detected events per unit time for the subject.

In some examples, only part of the non-fall event probability distribution is used in the calculation to determine the threshold value corresponding to a predetermined false alarm rate. This part may be determined using any of the previously described methods. In this way, a threshold value set by integrating the non-fall event probability distribution may yield comparable detection accuracy to a threshold value determined from the likelihood ratio test.

Other examples may include obtaining a predetermined threshold value, e.g. a standard threshold value used in the art.

FIG. **6** illustrates a fall detection system **600** for detecting a fall of a subject, including one or more sensors **610** and a processing system **620**, according to an implementation of the disclosure. The processing system **620** is, itself, an implementation of the disclosure.

The one or more sensors **610** are configured to obtain movement data **615** responsive to the subject's movement. The one or more sensors may include at least one of an accelerometer, a barometric sensor, a tilt switch, a vibration sensor and a gyroscope. The one or more sensors may be part of a wearable device, for example, a smartwatch or a pendant worn around the neck.

The processing system **620** includes one or more processors **621** adapted to receive movement data **615** from the one or more sensors **610**, and to detect events that occur in the

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movement data **615**. An event may be detected when a defined trigger parameter (which may be a combination of parameters of the movement data) changes by more than a predetermined amount or when a value of the trigger parameter breaches a predetermined threshold, as previously described. Other methods would be apparent to the skilled person.

Event information **622** from events that occur during a predetermined data collection period is stored in a memory **623**. This event information includes the values of a one or more parameters that may (together or individually) indicate the probability of a fall at the time of the event for each detected event, such as height, height change, orientation, orientation change, air pressure change, velocity, acceleration, duration and impact.

At the end of the predetermined data collection period, the one or more processors **621** obtain the event information **622** from the memory **623**, and generate a non-fall event probability distribution by calculating a probability distribution from the event information. In particular, a non-fall event probability distribution is calculated for a combination of the one or more parameters for which values are provided in the event information. Preferably, the one or more parameters includes a plurality of parameters.

The one or more processors **621** obtain a fall event probability distribution for the combination of the one or more parameters using the non-fall event probability distribution.

The one or more processors **621** also obtain a threshold value for the fall event probability distribution. This threshold value can be used to distinguish between fall events and non-fall events. The threshold value may be stored in the memory **623**.

In some examples, the one or more processors are adapted to process the non-fall event probability distribution and the fall event probability distribution to determine a threshold value **624** for the fall event probability distribution, such that the threshold value **624** may be used to distinguish between fall events and non-fall events. The non-fall event probability distribution and the fall event probability distribution may, for example, be processed using any of the described methods. The determined threshold value **624** is stored in the memory **623**.

The memory **623** may store a predetermined estimation of a fall event probability distribution **625** which may be used by the one or more processors **621** when determining the threshold value **624** and/or the fall event probability distribution.

For events that are detected after the predetermined data collection period has elapsed, the one or more processors **621** may compare the fall event probability for the value of the combination of the one or more parameters with the threshold value **624** and classify the event as a fall event or a non-fall event on the basis of this comparison.

The fall detection system **600** may also include a user interface **630** configured to provide a cancel function **635**. The cancel function **635** may be configured to re-classify a detected fall event as a non-fall event when activated by the subject.

In some examples, the threshold value **624** corresponds to a predetermined false alarm rate. A false alarm probability can be determined as the quotient between the predetermined false alarm rate and a trigger rate, where the trigger rate is defined as the number of detected events per unit time for the subject. The false alarm rate and/or the false alarm probability may be sent to the user interface **630** to inform the subject of the accuracy of the system.

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In some implementations, the user interface **630** is configured to receive a user instruction to disable the cancel function, should the subject consider the system to be accurate enough that a cancel function is not required. In other implementations, the processing system **620** may automatically disable the cancel function if the determined false alarm probability is below a predetermined value, for example, 0.001 or 0.00001.

It will be understood that the disclosed methods are computer-implemented methods. As such, there is also proposed a concept of a computer program including code means for implementing any described method when said program is run on a processing system.

The skilled person would be readily capable of developing a processor for carrying out any herein described method. Thus, each step of a flow chart may represent a different action performed by a processor, and may be performed by a respective module of the processing processor.

As discussed above, the system makes use of a processor to perform the data processing. The processor can be implemented in numerous ways, with software and/or hardware, to perform the various functions required. The processor typically employs one or more microprocessors that may be programmed using software (e.g. microcode) to perform the required functions. The processor may be implemented as a combination of dedicated hardware to perform some functions and one or more programmed microprocessors and associated circuitry to perform other functions.

Examples of circuitry that may be employed in various examples of the present disclosure include, but are not limited to, conventional microprocessors, application specific integrated circuits (ASICs), and field-programmable gate arrays (FPGAs).

In various implementations, the processor may be associated with one or more storage media such as volatile and non-volatile computer memory such as RAM, PROM, EPROM, and EEPROM. The storage media may be encoded with one or more programs that, when executed on one or more processors and/or controllers, perform the required functions. Various storage media may be fixed within a processor or controller or may be transportable, such that the one or more programs stored thereon can be loaded into a processor.

Variations to the disclosed implementations/examples can be understood and effected by those skilled in the art in practicing the claimed disclosure, from a study of the drawings, the disclosure and the appended claims. In the claims, the word "comprising" does not exclude other elements or steps, and the indefinite article "a" or "an" does not exclude a plurality. A single processor or other unit may fulfill the functions of several items recited in the claims. The mere fact that certain measures are recited in mutually different dependent claims does not indicate that a combination of these measures cannot be used to advantage. A computer program may be stored/distributed on a suitable medium, such as an optical storage medium or a solid-state medium supplied together with or as part of other hardware, but may also be distributed in other forms, such as via the Internet or other wired or wireless telecommunication systems. If the term "adapted to" is used in the claims or description, it is noted that the term "adapted to" is intended to be equivalent to the term "configured to". Any reference signs in the claims should not be construed as limiting the scope.

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What is claimed is:

1. A computer-implemented method executed on data processing hardware that causes the data processing hardware to perform operations comprising:

obtaining movement data representing movement of a subject over a predetermined data collection period, the movement data comprising one or more parameters, each respective parameter comprising a corresponding value;

detecting an event based on the movement data;

determining a non-fall event probability distribution based on a combination of the one or more parameters and the corresponding values, the non-fall event probability distribution indicating a probability of a non-fall event;

obtaining a threshold value based on the non-fall event probability distribution, the threshold value distinguishing between the non-fall event and a fall event; and

configuring an automatic fall detector with the threshold value, the automatic fall detector configured to use the threshold value to distinguish between the fall event and the non-fall event for subsequently obtained movement data.

2. The method of claim 1, wherein the operations further comprise:

for each respective parameter of the one or more parameters, determining a corresponding parameter specific probability distribution,

wherein determining the non-fall event probability distribution is further based on a product of each corresponding parameter specific probability distribution.

3. The method of claim 1, wherein obtaining the threshold value based on the non-fall event probability distribution comprises determining a difference between the non-fall event probability distribution and a predicted fall event probability distribution.

4. The method of claim 1, wherein obtaining the threshold value based on the non-fall event probability distribution comprises determining an inverse or a complement of the non-fall event probability distribution.

5. The method of claim 1, wherein obtaining the threshold value based on the non-fall event probability distribution comprises dividing a predetermined estimation of a fall event probability distribution by the non-fall event probability distribution.

6. The method of claim 1, wherein the operations further comprise determining, based on a predetermined estimation of a fall event probability, the combination of the one or more parameters.

7. The method of claim 6, wherein determining the combination of the one or more parameters comprises determining that the predetermined estimation of the fall event probability satisfies a threshold value.

8. The method of claim 6, wherein determining the combination of the one or more parameters comprises identifying a group of the one or more parameters having the greatest corresponding values.

9. The method of claim 1, wherein obtaining the threshold value comprises processing the non-fall event probability distribution and a fall event probability distribution.

10. The method of claim 1, wherein obtaining the threshold value is further based on a false alarm rate representing a rate of incorrectly classified non-fall events.

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11. A system comprising:

data processing hardware; and

memory hardware in communication with the data processing hardware, the memory hardware storing instructions that when executed on the data processing hardware cause the data processing hardware to perform operations comprising:

obtaining movement data representing movement of a subject over a predetermined data collection period, the movement data comprising one or more parameters, each respective parameter comprising a corresponding value;

detecting an event based on the movement data;

determining a non-fall event probability distribution based on a combination of the one or more parameters and the corresponding values, the non-fall event probability distribution indicating a probability of a non-fall event;

obtaining a threshold value based on the non-fall event probability distribution, the threshold value distinguishing between the non-fall event and a fall event; and

configuring an automatic fall detector with the threshold value, the automatic fall detector configured to use the threshold value to distinguish between the fall event and the non-fall event for subsequently obtained movement data.

12. The system of claim 11, wherein the operations further comprise:

for each respective parameter of the one or more parameters, determining a corresponding parameter specific probability distribution,

wherein determining the non-fall event probability distribution is further based on a product of each corresponding parameter specific probability distribution.

13. The system of claim 11, wherein obtaining the threshold value based on the non-fall event probability distribution comprises determining a difference between the non-fall event probability distribution and a predicted fall event probability distribution.

14. The system of claim 11, wherein obtaining the threshold value based on the non-fall event probability distribution comprises determining an inverse or a complement of the non-fall event probability distribution.

15. The system of claim 11, wherein obtaining the threshold value based on the non-fall event probability distribution comprises dividing a predetermined estimation of a fall event probability distribution by the non-fall event probability distribution.

16. The system of claim 11, wherein the operations further comprise determining, based on a predetermined estimation of a fall event probability, the combination of the one or more parameters.

17. The system of claim 16, wherein determining the combination of the one or more parameters comprises determining that the predetermined estimation of the fall event probability satisfies a threshold value.

18. The system of claim 16, wherein determining the combination of the one or more parameters comprises identifying a group of the one or more parameters having the greatest corresponding values.

19. The system of claim 11, wherein obtaining the threshold value comprises processing the non-fall event probability distribution and a fall event probability distribution.

20. The system of claim 11, wherein obtaining the threshold value is further based on a false alarm rate representing a rate of incorrectly classified non-fall events.

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